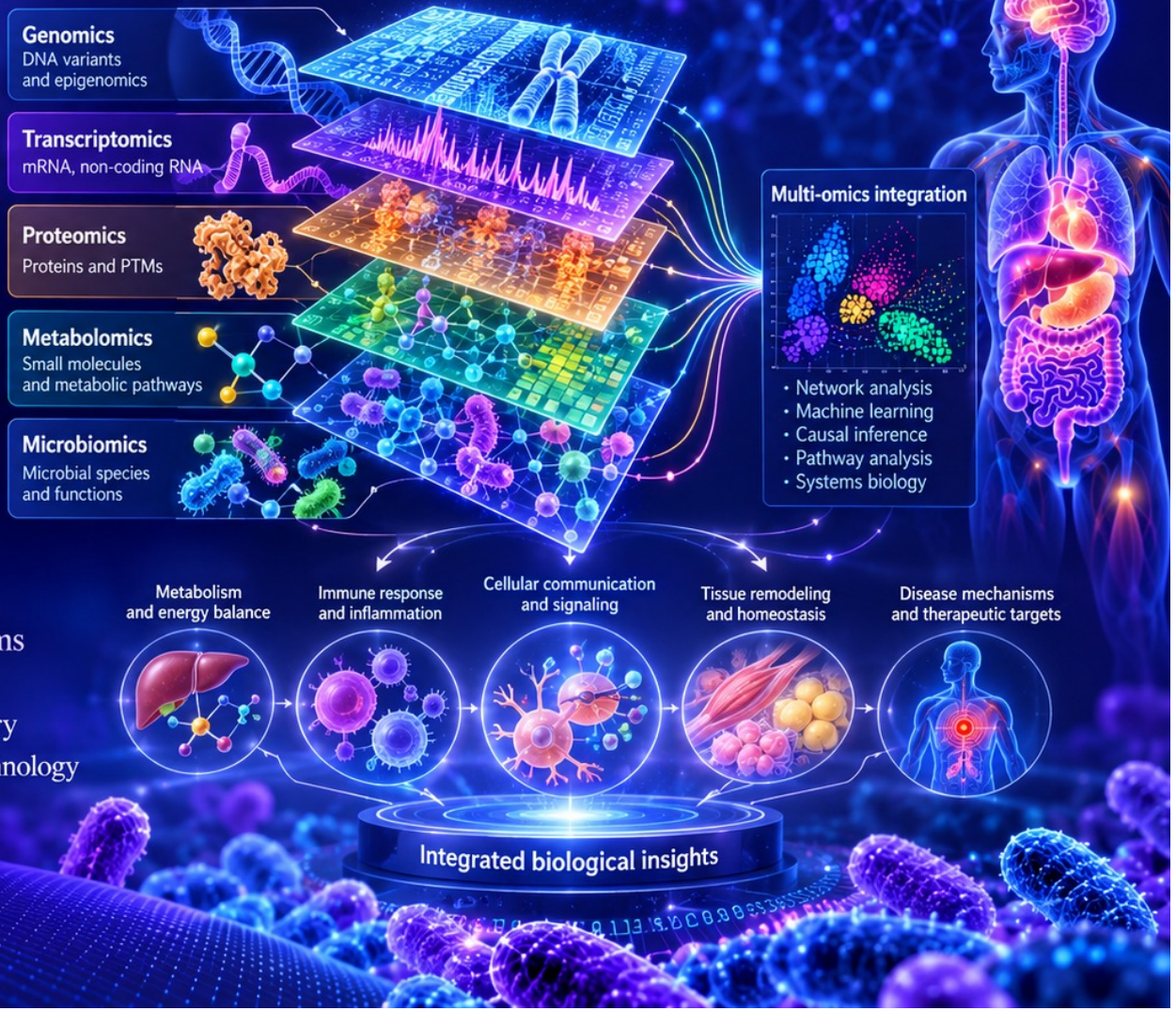
Week 9

Multi-Omics Integration – From Molecular Layers to Biological Insights

Integrating genomics, transcriptomics, microbiomics, proteomics and metabolomics to understand host biology and disease mechanisms

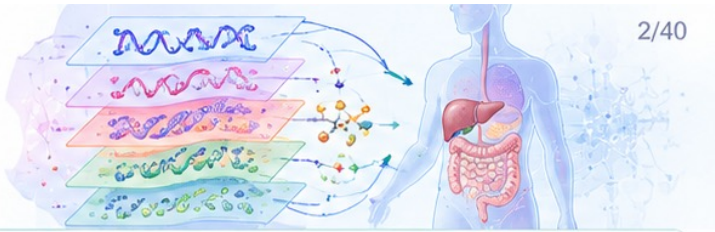
Bioinformatics: From Multi-Omics Data to Discovery
Dr. Liwei Xie | Shenzhen University of Advanced Technology

Two 40-Minute Sessions | Fall 2026



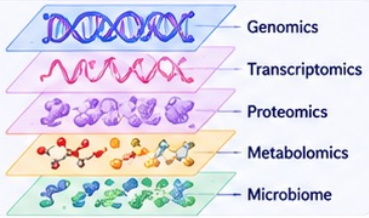
Week 9 Multi-Omics Integration

From Molecular Layers to Biological Mechanisms and Hypotheses



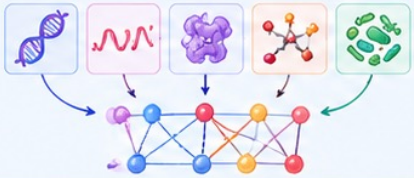
1 Why Integrate Multi-Omics?

- Limitations of single-omics studies
- Complementary information across layers
- From data integration to biological insight



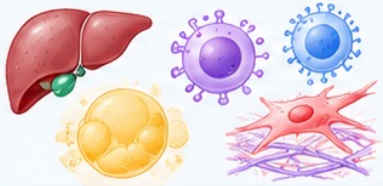
2 Integration Strategies

- Horizontal vs vertical integration
- Early vs late vs joint integration
- Network-based and machine learning approaches



3 Host Functions as a Framework

- Metabolism and energy homeostasis
- Immune response and inflammation
- Tissue remodeling and fibrosis
- Cell–cell communication



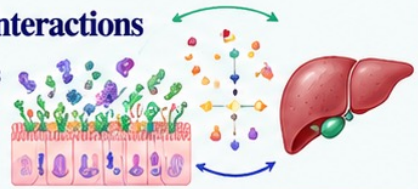
4 Pathway and Functional Analysis

- GO, KEGG and Reactome annotation
- ORA vs GSEA: principles and interpretation
- From gene lists to biological functions



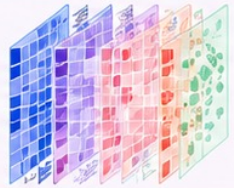
5 Microbiome–Metabolite–Host Interactions

- Microbial functions and key metabolites
- Metabolite–host signaling pathways
- Gut–liver and gut–muscle axis



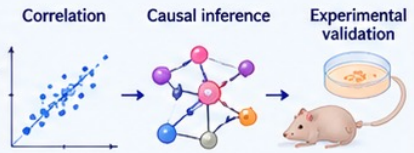
6 Integrative Analysis Across Omics

- Cross-omics correlation and network analysis
- Multi-omics concordance and evidence integration
- Cell type and tissue context



7 From Association to Mechanism

- Causality inference and perturbation analysis
- Prioritizing key targets and pathways
- Experimental validation strategies



8 AI-Driven Multi-Omics Analysis

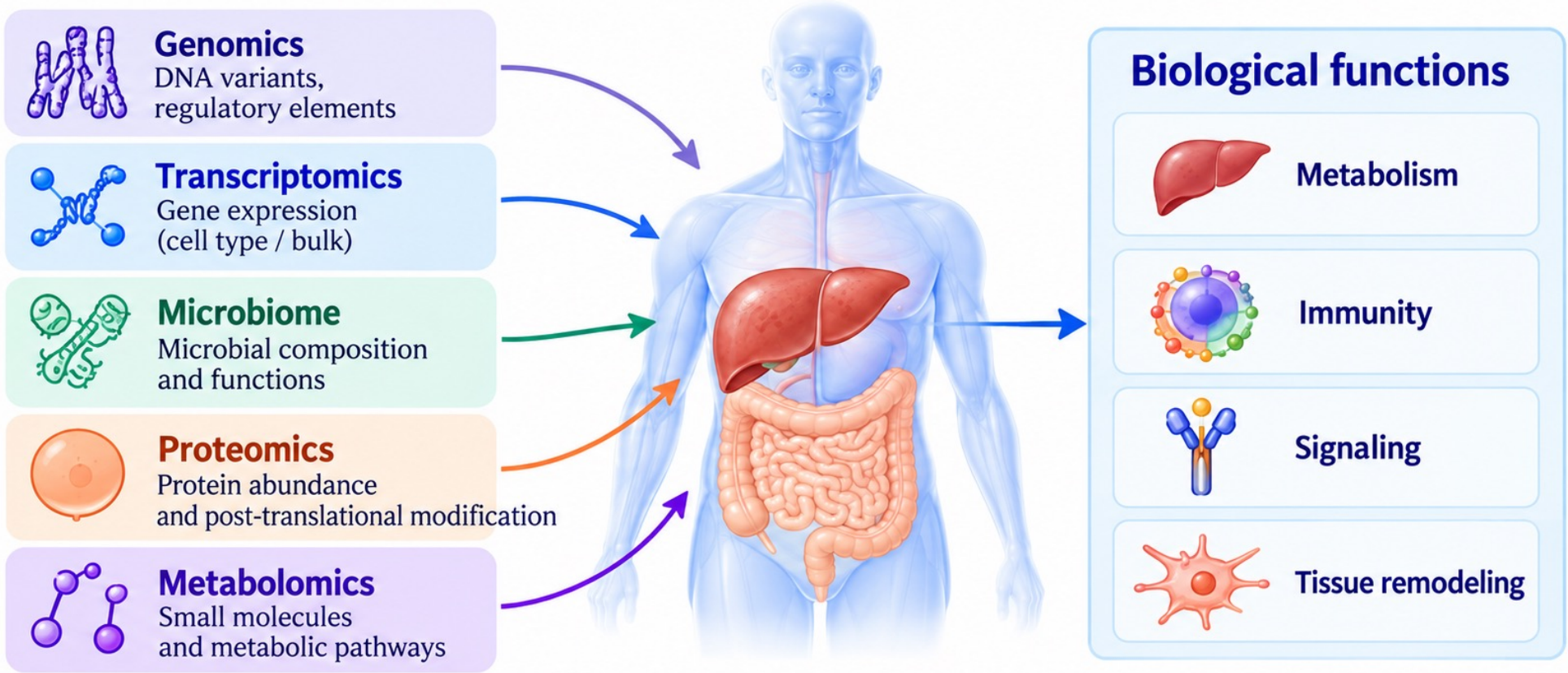
- AI tools and agents for integration analysis
- Automating omics interpretation
- From data to hypothesis generation



Take-home message: Multi-omics integration connects molecular layers to host functions and generates testable hypotheses.

From Multi-Omics to Biological Discovery

Integrating multiple molecular layers to understand host biology and disease.

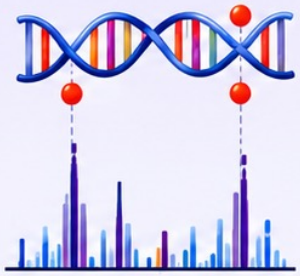


Take-home message: Biological discovery requires connecting molecular layers, not analyzing each in isolation.

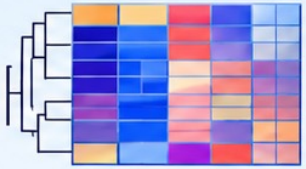
Why Is One Omics Layer Not Enough?

Each omics layer provides a different view of the same biological system.

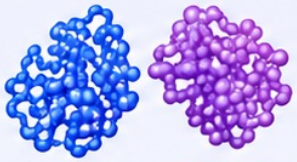
Genomics
Risk variants



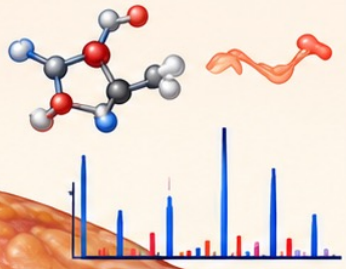
Transcriptomics
Gene expression



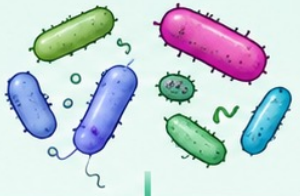
Proteomics
Protein changes

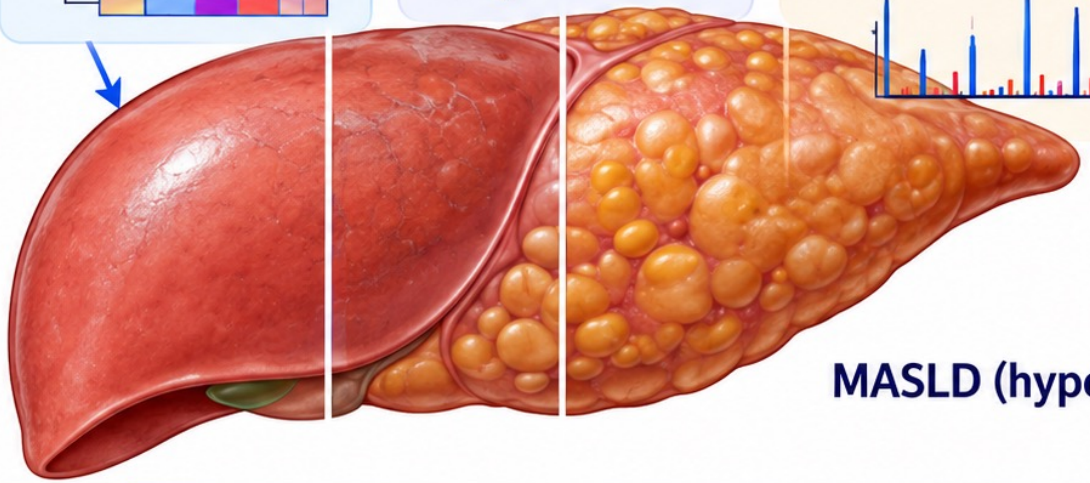


Metabolomics
Metabolic alterations



Microbiome
Gut–liver crosstalk





Normal

MASLD (hypothetical)

Take-home message: Every omics layer captures only part of the biological truth.

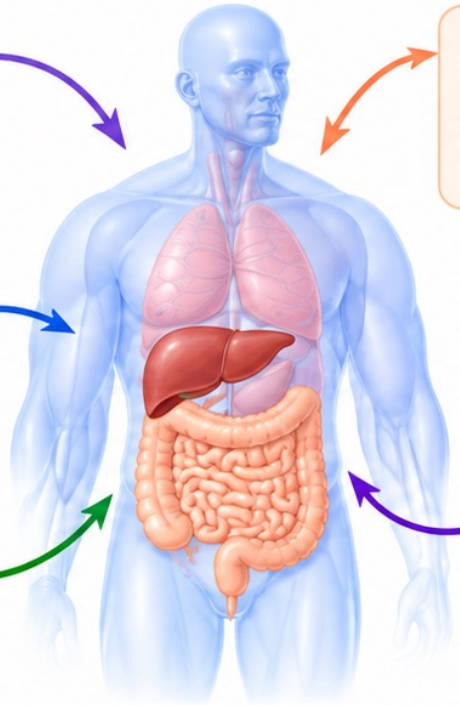
Five Molecular Layers, One Host

Multiple molecular measurements converge on shared host biological functions.

Genomics
DNA sequence
regulatory variants

Transcriptomics
RNA expression
bulk / single-cell

Microbiome
microbial species
and functions



Proteomics
protein abundance
and modification

Metabolomics
small molecules
and pathways

Host functions

- Metabolism
- Immunity
- Signaling
- Tissue remodeling

Take-home message: Host biological functions connect otherwise separate molecular data types.

What Do Different Omics Actually Measure?

Omics layers capture information at different molecular levels and scales.

Genome

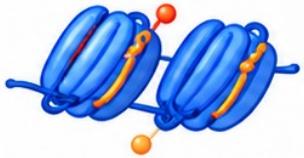
DNA sequence and genetic variants



- SNP
- Indel
- CNV
- SV

Epigenome

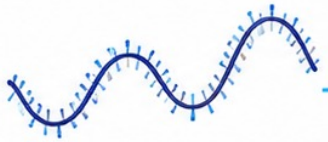
Chromatin state and regulation



- DNA methylation
- Chromatin accessibility
- Histone modification

Transcriptome

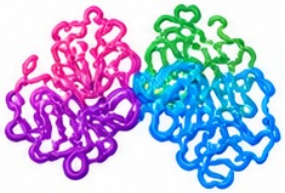
Gene expression and isoforms



- mRNA levels
- Alternative splicing
- Non-coding RNA

Proteome

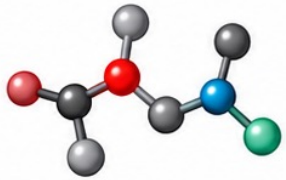
Protein abundance and activity



- Protein levels
- Post-translational modification
- Protein interaction

Metabolome

Small molecules and pathways



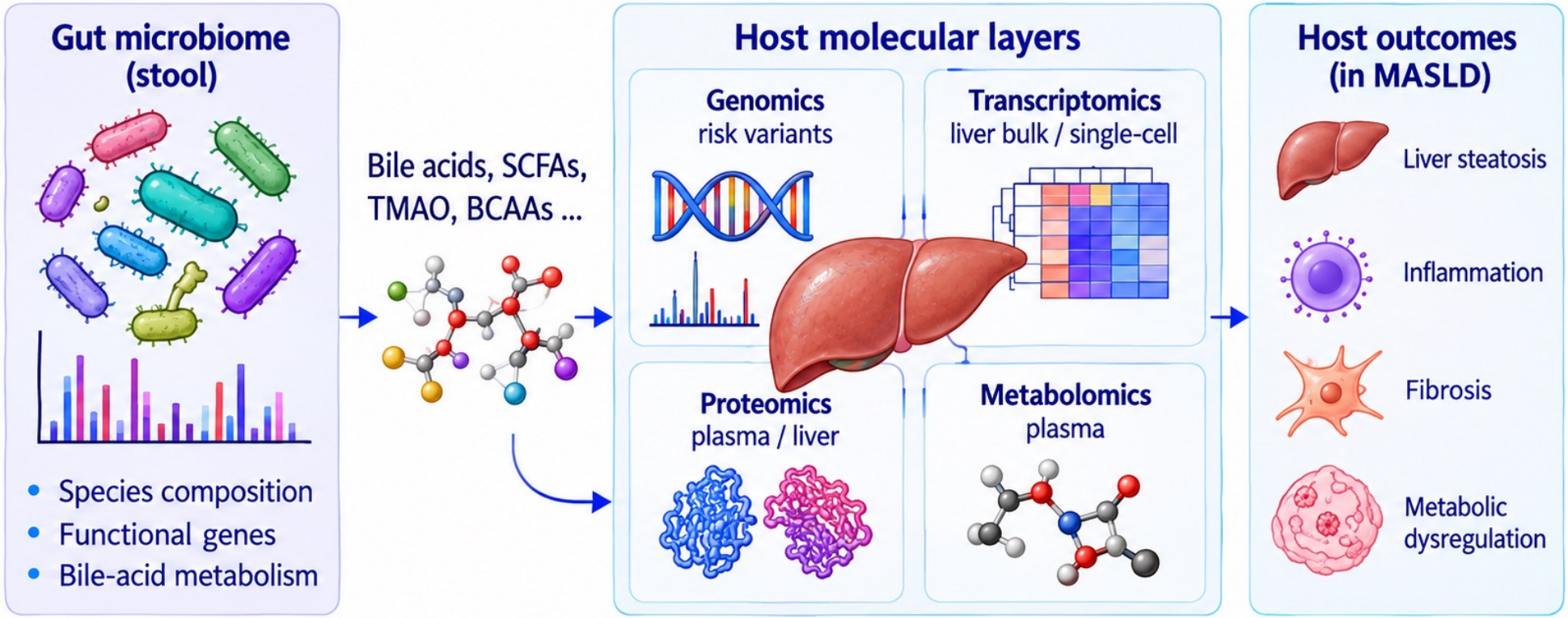
- Metabolite levels
- Metabolic flux
- Pathway activity
- Metabolite–protein interaction

More upstream (genetic potential) → More downstream (functional phenotype)

Take-home message: Different molecular measurements answer different biological questions.

One Scientific Question Across Five Layers

A running example: gut–liver MASLD study with matched multi-omics data.

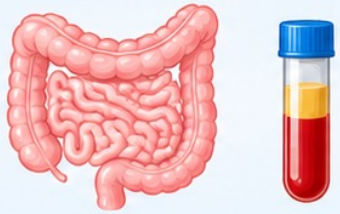


Take-home message: Define the biological question before selecting an integration method.

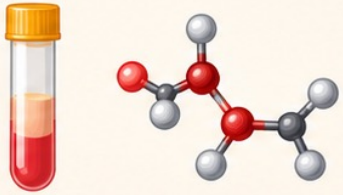
Sample Matching Comes Before Integration

All omics layers must be linked by the same subjects, visits and biospecimens.

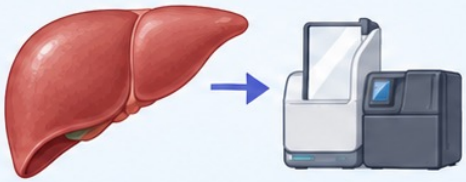
Stool microbiome



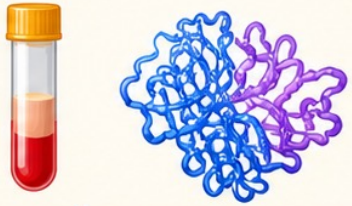
Plasma metabolomics



Liver RNA-seq bulk / single-cell



Plasma / liver proteomics



Master Sample Manifest

sample_id	subject_id	visit	stool	plasma	liver	microbiome	metabolome	RNA	proteome
S01	V01	✓	✓	✓	✓	✓	✓	✓	✓
S02	V01	✓	✓	✓	✓	✓	✓	✗	✓
S03	V02	✓	✓	✓	✗	✗	✓	✗	✗
…	…	…	…	…	…	…	…	…	…

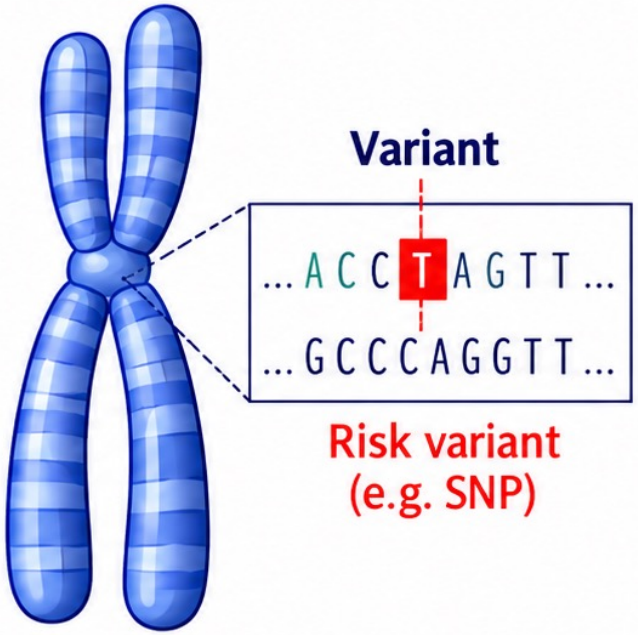
✓ Available

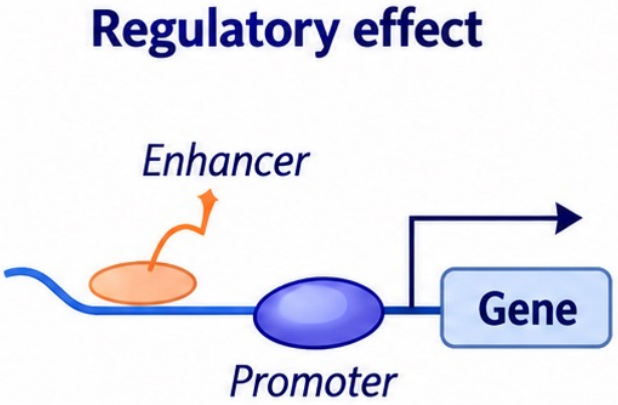
✗ Missing

Take-home message: Incorrect sample pairing creates artificial biological relationships.

Genomics: The Host Regulatory Blueprint

Genetic variants can affect gene regulation and downstream molecular changes.





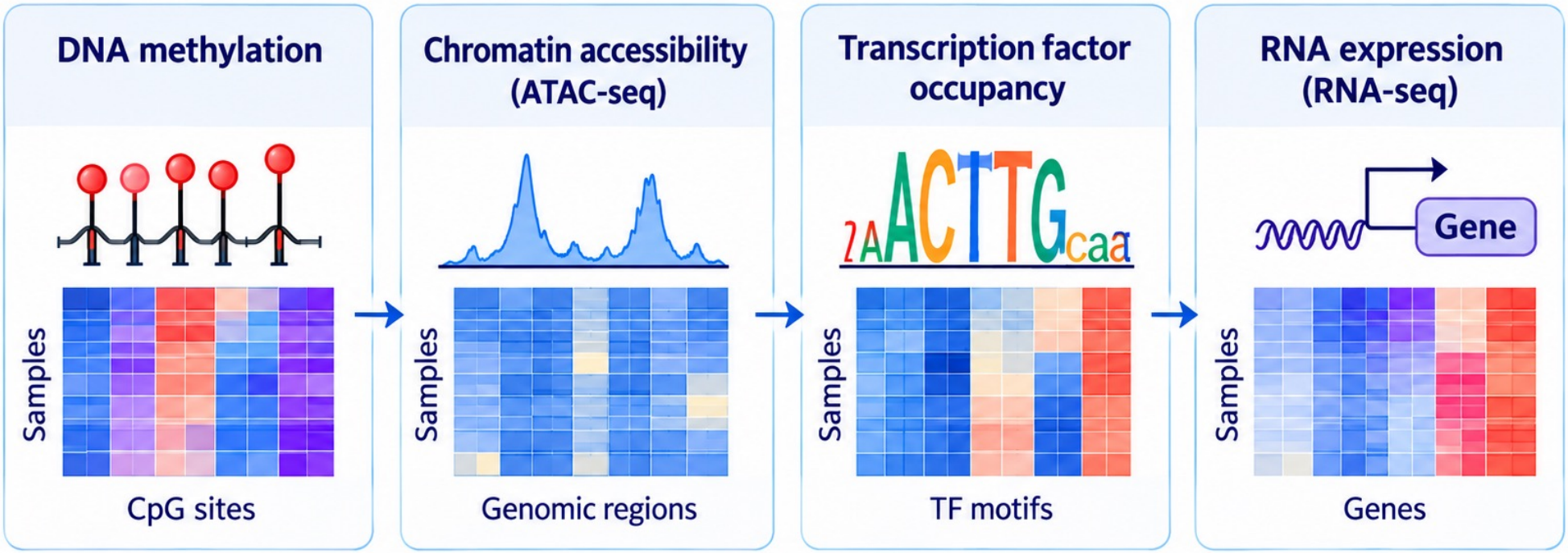
Possible consequences

- Increased/decreased transcription
- Altered chromatin state
- Effects on protein function
- Disease susceptibility

Take-home message: Genomics identifies potential regulatory mechanisms, not their activity.

From Chromatin Regulation to Gene Expression

Integrating epigenomic and transcriptomic data links regulatory control to expression.

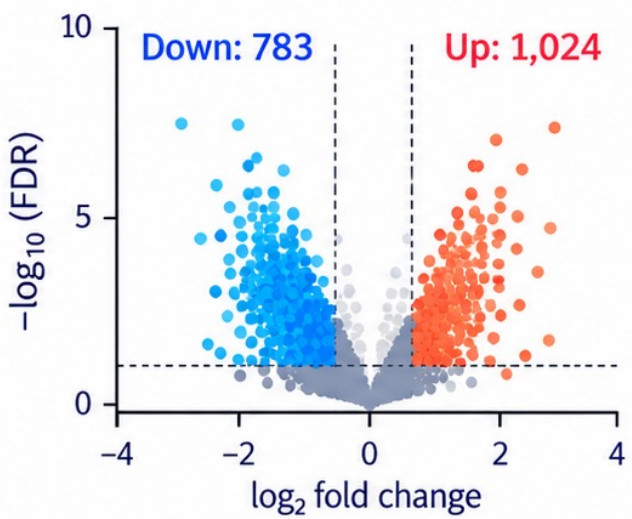


Take-home message: Integrating regulatory and expression data links potential control to cellular responses.

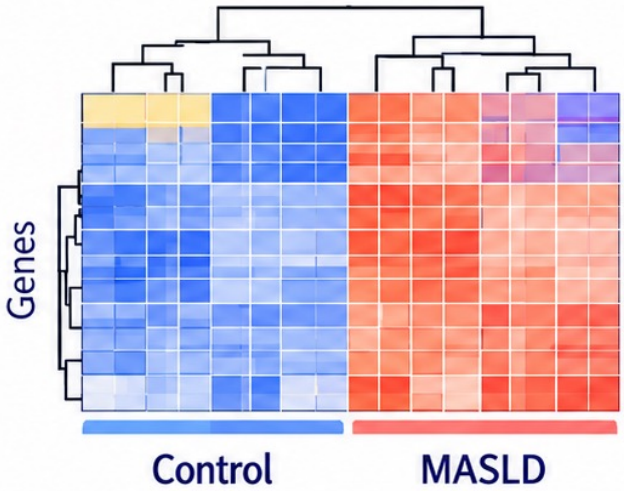
Transcriptomics: What Is the Host Responding To?

RNA expression reveals coordinated host responses and pathway changes.

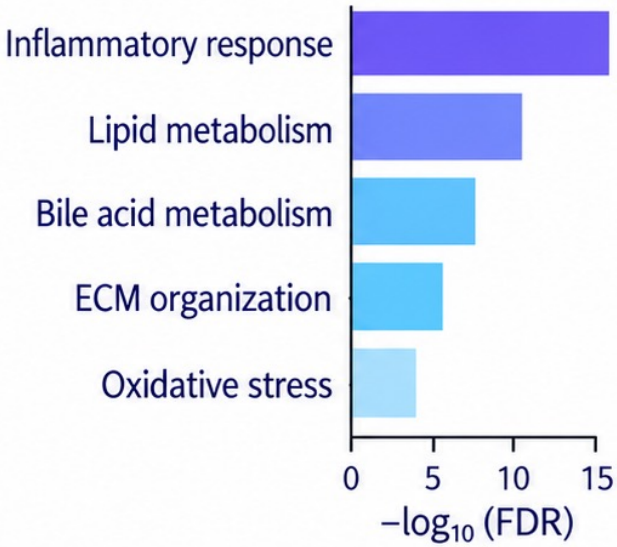
Differential expression
(MASLD vs control)



Expression heatmap
(top variable genes)



Enriched pathways

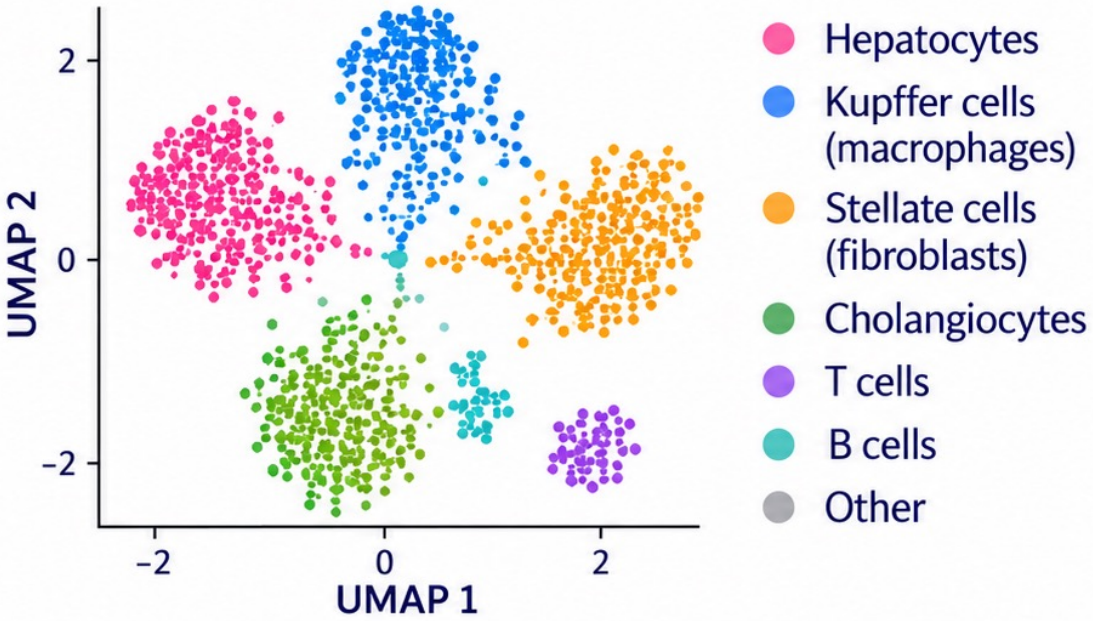


Take-home message: Transcriptomics reveals coordinated host expression responses.

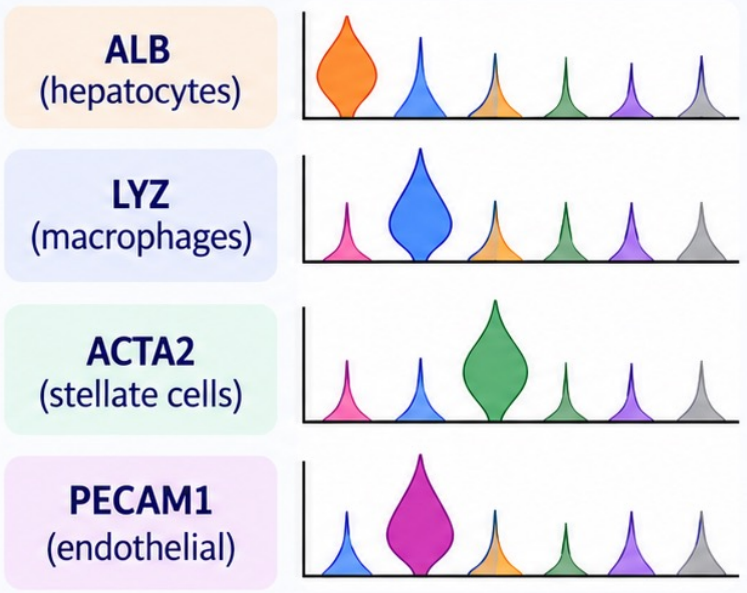
Single-Cell Omics: Which Cells Respond?

Cell-type resolution identifies where molecular changes occur in the tissue.

Liver single-cell RNA-seq (UMAP)



Example marker genes



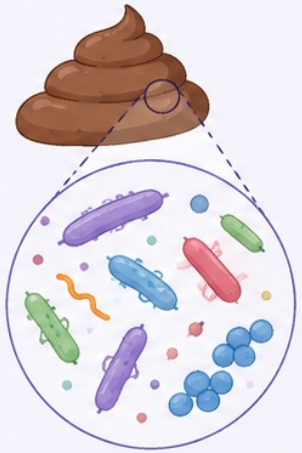
Take-home message: Cell-type resolution localizes tissue-level molecular signals.

Microbiomics: Which Functions Might Microbes Provide?

Taxonomy, functional potential and microbial metabolites offer different levels of evidence.

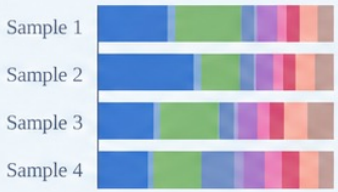
STOOL MICROBIOME

Complex microbial community



TAXONOMY

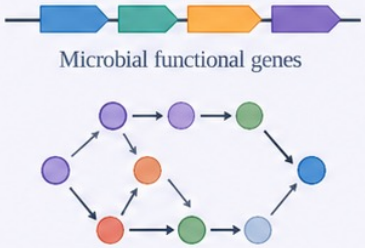
Species composition from 16S or shotgun metagenomics



- *Bacteroides*
- *Faecalibacterium*
- *Prevotella*
- *Ruminococcus*
- *Akkermansia*
- Others

FUNCTIONAL POTENTIAL

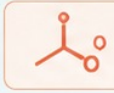
Microbial genes and metabolic pathways (from metagenomics)



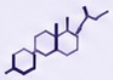
Metabolic pathways (e.g., SCFA synthesis, bile acid modification)

METABOLITE OUTPUT

Microbial metabolites measured in stool, plasma or other biospecimens



SCFAs
(acetate, propionate, butyrate)



Bile acids
(primary/secondary)



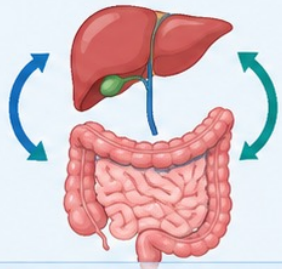
TMAO
(from choline, carnitine)



BCAAs
(leucine, isoleucine, valine)

HOST CONSEQUENCES

Microbial metabolites act on the gut–liver axis and influence host physiology.



- Liver metabolism
- Immune modulation
- Barrier function
- Inflammation
- Metabolic health

Measured functions
Which microbes are present?

Predicted functions
What can they do and what do they produce?

Species ≠ function
Different species can share functions, and one species can have multiple functions.

TAKE-HOME MESSAGE Microbial taxonomy, functional potential and metabolite output provide complementary evidence.

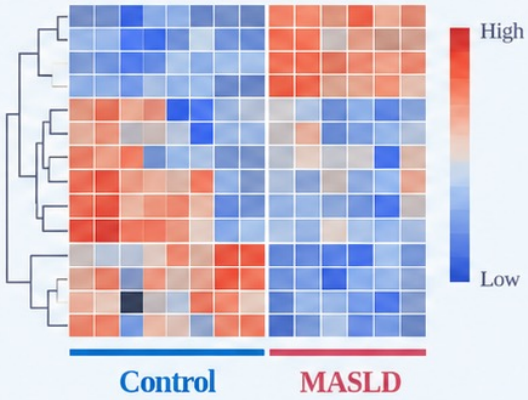
Proteomics: Do RNA Changes Reach Proteins?

RNA abundance, protein abundance and protein activity are related, but not equivalent.

RNA EXPRESSION CHANGES

Transcription

Genes are transcribed into RNA.

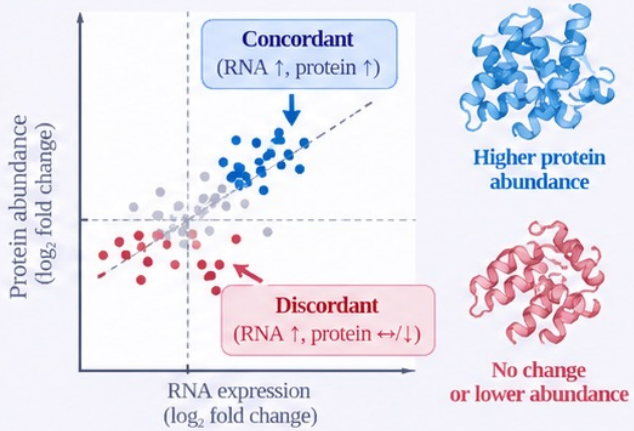


Differential gene expression (RNA-seq)

PROTEIN ABUNDANCE CHANGES

Translation / protein abundance

RNAs are translated into proteins.



Not all RNA changes appear at the protein level.

FUNCTIONAL PROTEIN LAYER

Activity / post-translational modification

Proteins are modified and activated, affecting cellular functions.

Post-translational modifications



- Phosphorylation
- Acetylation
- Ubiquitination
- Other modifications

Signaling activity



- Kinase activation
- Signaling pathways
- Protein–protein interactions

Functional outcome



- Metabolism
- Immunity
- Cell communication
- Tissue remodeling

TAKE-HOME MESSAGE RNA and protein changes may agree, disagree, or differ in functional meaning.

Metabolomics: The Biochemical Phenotype

Metabolites provide chemical evidence close to phenotype, but source and flux still require interpretation.

METABOLITE SOURCES



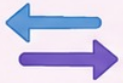
Host metabolism
Endogenous metabolites from host tissues (e.g., liver, immune cells, energy metabolism).



Microbiome metabolism
Microbial metabolites (e.g., SCFAs, TMA, indoles).

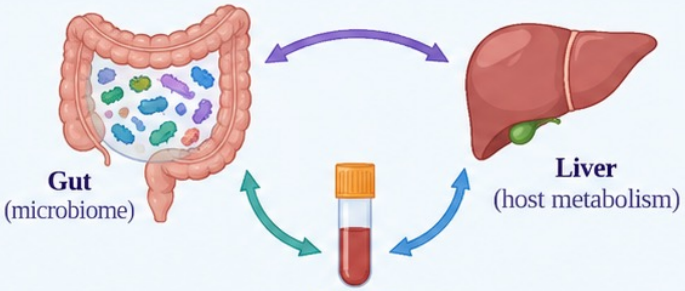


Dietary intake
Food-derived metabolites (e.g., polyphenols, amino acids, choline, lipids).

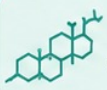


Co-metabolism
Host–microbe co-metabolism generates shared metabolites (e.g., secondary bile acids).

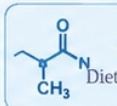
THE BIOCHEMICAL PHENOTYPE



REPRESENTATIVE METABOLITE CLASSES



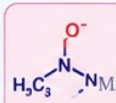
Bile acids
Host/microbe



BCAAs
Diet/host/microbe



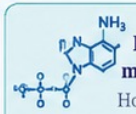
SCFAs
Microbiome



TMAO
Microbiome/diet



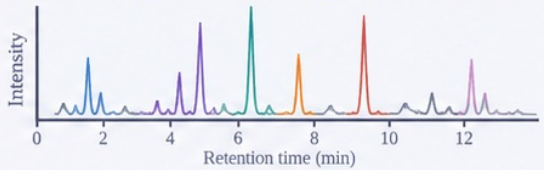
Lipids
Host/diet/microbe



Energy metabolites
Host/microbe

METABOLOMICS ANALYSIS

LC-MS METABOLITE PROFILE



METABOLITE NETWORK



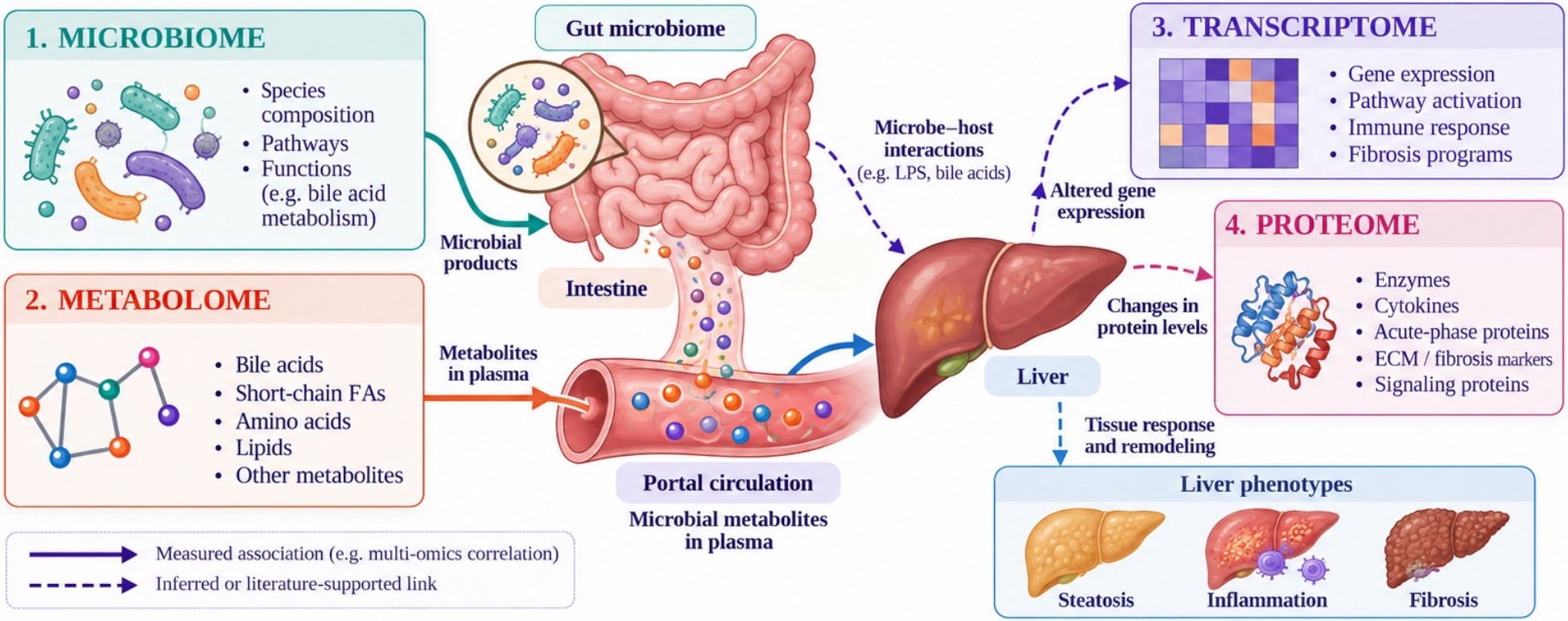
INTERPRETATION CONSIDERATIONS

Source	Host, microbiome, diet, or co-metabolism?
Abundance	Relative or absolute changes in metabolite levels.
Pathway context	Which metabolic pathways are involved?
Flux not directly measured	Concentrations do not reveal direction or rate of flow.

TAKE-HOME MESSAGE Metabolomics captures biochemical phenotype, but metabolite source and pathway direction need validation.

The Gut–Liver Axis Connects Molecular Layers

Microbial products, host responses and tissue phenotypes can be linked through gut–plasma–liver biology.



TAKE-HOME MESSAGE The gut–liver axis provides a biological framework for multi-omics integration.

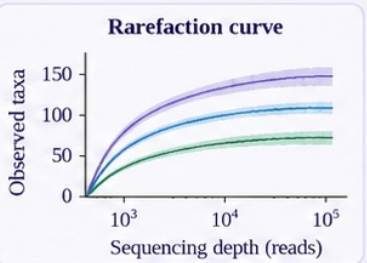
Different Omics Need Different Preprocessing

Quality control, normalization and missing-data handling must respect assay-specific biology.

MICROBIOME



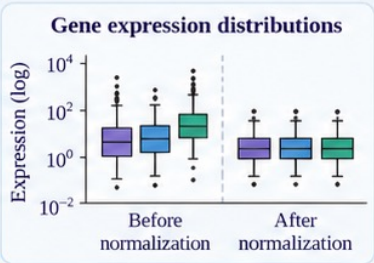
- Filter low-abundance taxa (e.g., rare < 0.1%).
- Handle compositional data (e.g., CLR, ALR).
- Remove contaminants and low-quality samples.



RNA-SEQ



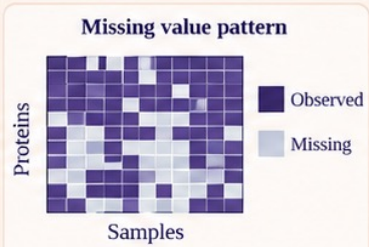
- Filter lowly expressed genes (e.g., CPM/TPM).
- Normalize for library size (e.g., TMM, DESeq2).
- Correct batch effects (e.g., ComBat, Harmony).



PROTEOMICS



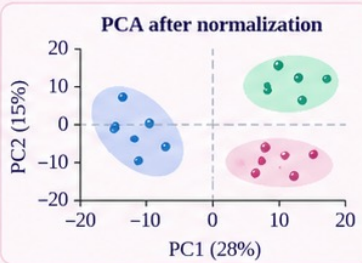
- Filter proteins with high missingness (e.g., > 50%).
- Impute missing values (e.g., KNN, low-rank).
- Normalize across samples (e.g., median, quantile).



METABOLOMICS



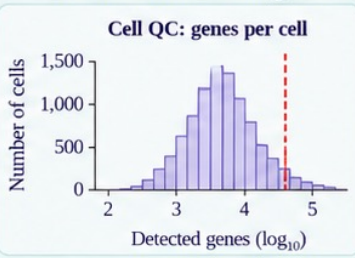
- Filter low-quality peaks (e.g., noise, low signal).
- Annotate features (e.g., m/z to metabolites).
- Normalize (e.g., PQN, median scaling).
- Log-transform and scale.



SINGLE-CELL / PSEUDOBULK



- Filter low-quality cells (e.g., low genes, high mito).
- Remove doublets (e.g., Scrublet).
- Normalize (e.g., log1p, SCTransform).
- Aggregate to pseudobulk for multi-omics integration.



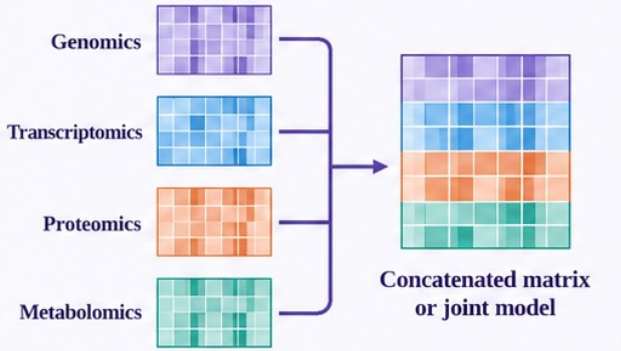
TAKE-HOME MESSAGE Valid integration begins with assay-specific preprocessing and quality control.

Three Strategies for Multi-Omics Integration

Early, intermediate and late integration answer different types of questions.

EARLY INTEGRATION

Concatenate or jointly model matched matrices.



BEST USE

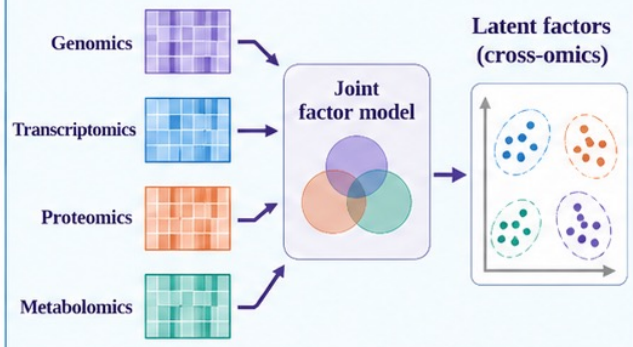
- Matched samples
- Model cross-omics dependencies

CONSIDERATIONS

- Requires same samples
- High dimensionality
- Different data scales

INTERMEDIATE INTEGRATION

Learn latent factors or cross-omics components.



BEST USE

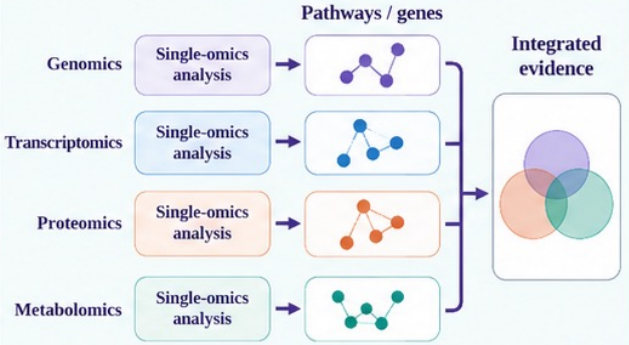
- Shared latent biology
- Uncover multi-omics patterns
- Dimension reduction

CONSIDERATIONS

- Model assumptions
- Choice of method (e.g., MOFA, iCluster)
- Interpretability

LATE INTEGRATION

Analyze each omics layer separately, then compare results.



BEST USE

- Pathway-level comparison
- Different sample sets or study types
- Validate consistent signals

CONSIDERATIONS

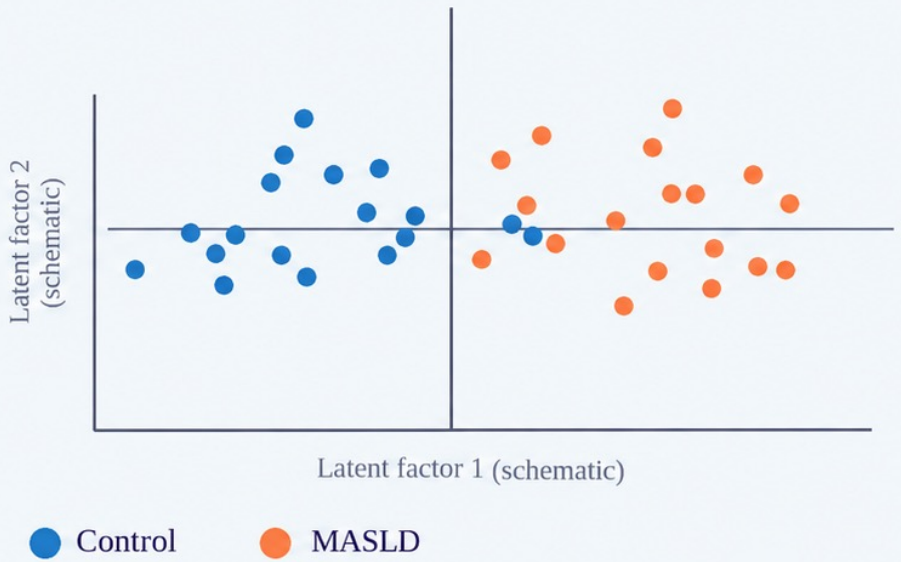
- May miss direct cross-omics relationships
- Dependent on single-omics analyses
- Integration at results level

TAKE-HOME MESSAGE Choose the integration strategy that matches the study design and scientific question.

MOFA2 Reveals Shared Latent Biology

Latent factors summarize cross-omics variation, but interpretation still requires biological checks.

FACTOR SCORES



VARIANCE EXPLAINED ACROSS VIEWS



LOADING / PATHWAY INTERPRETATION

- Latent factors combine signals across omics layers.
- Factor loadings can highlight contributing features and pathways.
- Biological interpretation requires feature-level and pathway checks.

Illustrative teaching data — not estimates from a published MASLD cohort.

TAKE-HOME MESSAGE Latent factors summarize shared variation, but mechanism must be interpreted afterwards.

DIABLO Finds Cross-Omics Predictive Signatures

Supervised integration can classify outcomes, but evaluation must be truly held out.

MATCHED OMICS DATA

- Genomics
- Transcriptomics
- Microbiome
- Proteomics
- Metabolomics

Matched samples across molecular layers

DIABLO

Supervised multi-omics integration (mixOmics)



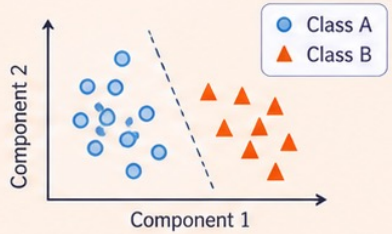
- Selects correlated features across omics layers
- Learns components maximizing outcome prediction
- Features are selected **within training folds only**

CROSS-OMICS SIGNATURE

- Genomics
- Transcriptomics
- Microbiome
- Proteomics
- Metabolomics

- Selected cross-omics signature
- Correlated features across layers
- Low-dimensional components for prediction

PREDICTIVE CLASSIFICATION



- Predict outcome for untouched samples
- Evaluate performance (BER / sensitivity / specificity)
- Use final model to classify new samples

NESTED VALIDATION (TRULY HELD-OUT EVALUATION)

Outer training fold
Normalize & filter
TRAIN ONLY

→

Inner cross-validation
Tune keepX + components
Repeated CV

→

Fit final fold model
Selected cross-omics signature

→

Outer test fold
Predict untouched samples
BER / sensitivity / specificity

→

External validation (if available)
Evaluate on independent cohort

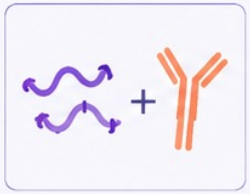
TAKE-HOME MESSAGE: Supervised integration is useful for prediction only when validated on untouched data.

Modern Integration: Match Method to Data

Different data geometries require different models and introduce different risks.

totalVI

Integrate RNA and protein in the same cells



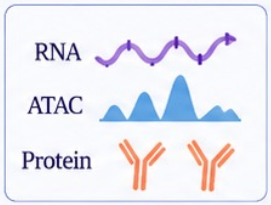
CITE-seq

Input geometry:
Paired RNA + protein (single cells)

Key caution:
Requires matched modalities (same cells).

MultiVI

Integrate multiple modalities in the same cells



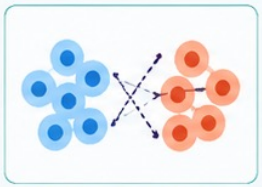
Multi-omics (same cells)

Input geometry:
Paired RNA + ATAC (+ protein)

Key caution:
Requires matched modalities (same cells).

GLUE

Align unpaired datasets using graph-based learning



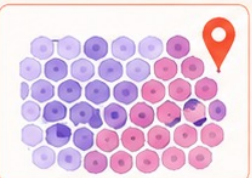
Unpaired datasets + graph structure

Input geometry:
Unpaired modalities + graph (e.g., kNN)

Key caution:
Performance depends on graph quality and appropriate anchors.

Spatial graph methods

Integrate spatial context with molecular profiles



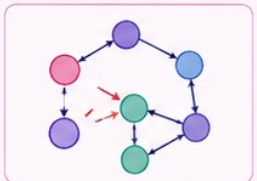
Spatial transcriptomics + spatial graph

Input geometry:
Spatial coordinates + molecular data

Key caution:
Requires accurate spatial information; sensitive to neighborhood definition.

COSMOS

Use mechanistic priors to guide integration



Mechanistic network priors

Input geometry:
Multi-omics + mechanism (e.g., pathways, regulatory networks)

Key caution:
Depends on quality and completeness of prior mechanistic knowledge.

Foundation models

Leverage large models for cross-study integration



Pretrained model → new datasets / tissues

Input geometry:
Transfer across studies, tissues, or species

Key caution:
Risk of out-of-distribution (OOD) generalization and batch-driven artifacts.

TAKE-HOME MESSAGE No method is universally best; match the model to the data geometry and question.

Can You Choose the Right Strategy?

Study design, sample pairing and the research question determine which conclusions are defensible.

SCENARIOS

1. Matched discovery cohort

Same subjects measured across multiple omics layers.



Identify shared structure across layers?

2. Matched outcome-prediction cohort

Same subjects with multiple omics and a clinical outcome.



Build a model to predict outcome or disease status?

3. Unmatched multi-cohort comparison

Different cohorts or data types with no shared samples.

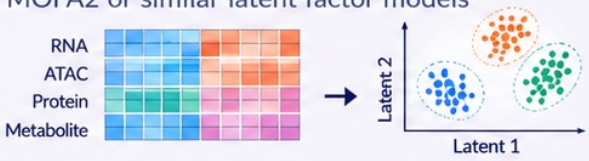


Compare pathways or biological themes across cohorts?

RECOMMENDED STRATEGY

Unsupervised multi-omics discovery

MOFA2 or similar latent factor models



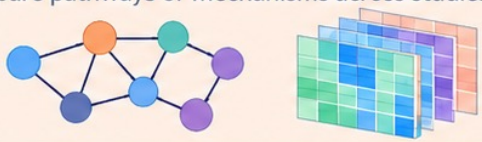
Supervised prediction and discrimination

DIABLO or supervised multi-omics methods with nested validation



Late evidence integration

Compare pathways or mechanisms across studies



SPECIAL GEOMETRY

Some data types or designs require tailored methods.

CITE-seq → totalVI



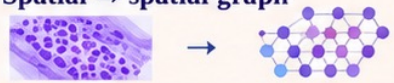
RNA + ATAC → MultiVI



Unpaired + graph → GLUE



Spatial → spatial graph



Mechanism model → COSMOS



TAKE-HOME MESSAGE

Method choice should follow design and biology—not software popularity.

From Molecular Data to Biological Functions

Multiple omics layers converge on shared host biological pathways.

Multi-omics data layers

Genomics
Genetic variants

Transcriptomics
Gene expression

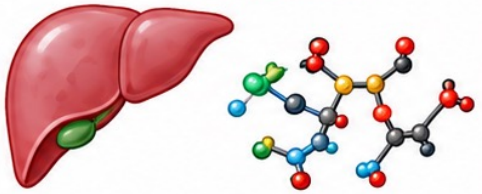
Proteomics
Protein abundance

Metabolomics
Small molecules

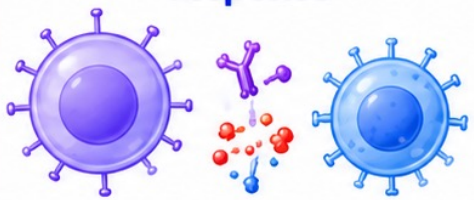
Microbiome
Microbial species and functions

Shared host pathway modules

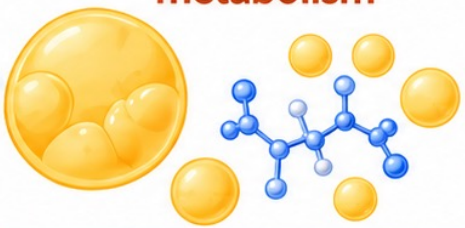
Bile acid metabolism



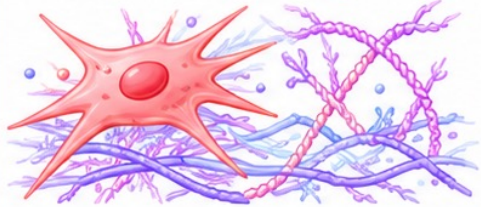
Immune response



Lipid metabolism



Tissue remodeling

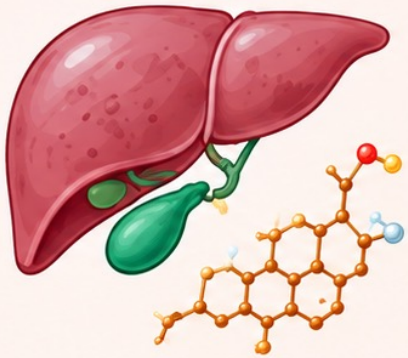


Take-home message: Host pathways provide a common language across molecular layers.

Four Host Functions Connect Multi-Omics

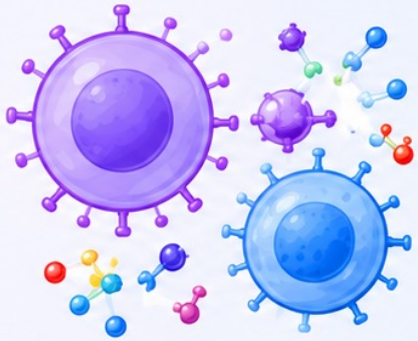
Key host functions organise multi-omics findings into testable questions.

Bile acid metabolism



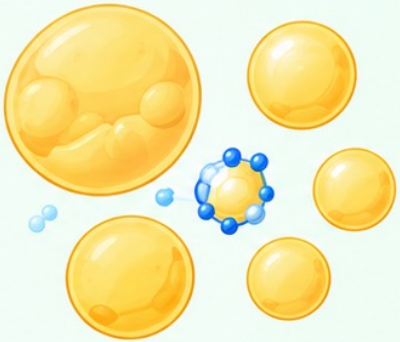
- Bile acid synthesis
- Bile acid signalling
- Microbiome conversion
- FXR/TGR5 pathways

Inflammation and immunity



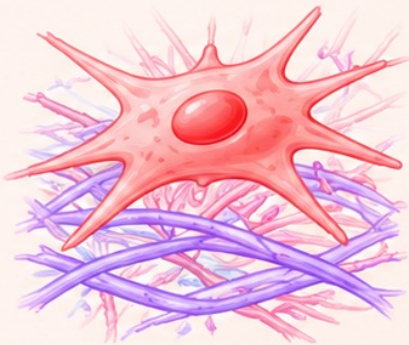
- Innate immune response
- Cytokine signalling
- Macrophage activation
- Inflammatory resolution

Lipid transport and metabolism



- Fatty acid metabolism
- Lipid transport
- VLDL/LDL/HDL
- Steatosis and lipotoxicity

ECM remodelling and fibrosis



- Stellate cell activation
- ECM deposition
- Tissue remodelling
- Fibrosis progression

Take-home message: Biological functions organise multi-omics findings into testable questions.

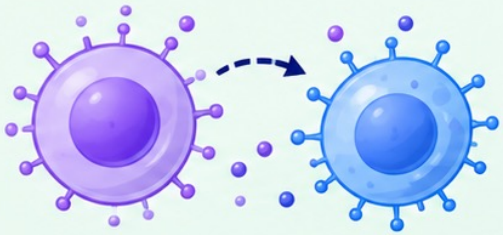
Gene Ontology: What Do Genes Do?

GO classifies gene functions into biological process, molecular function and cellular component.

Gene Ontology (GO)

Biological Process (BP)

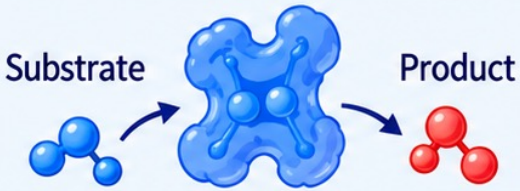
Biological objectives and pathways



- Immune response
- Lipid metabolism
- Inflammatory signalling
- Cell proliferation

Molecular Function (MF)

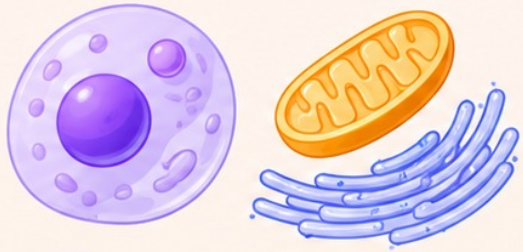
Molecular activities at the biochemical level



- Enzyme activity
- Receptor binding
- Transporter activity
- DNA / RNA binding

Cellular Component (CC)

Where gene products act in the cell



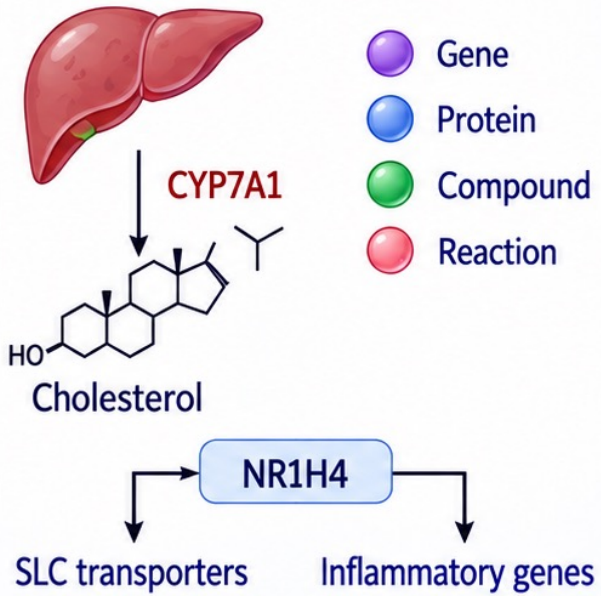
- Plasma membrane
- Mitochondrion
- Nucleus
- Extracellular region

Take-home message: GO describes annotated function, process and cellular location.

KEGG and Reactome: Mapping Biological Systems

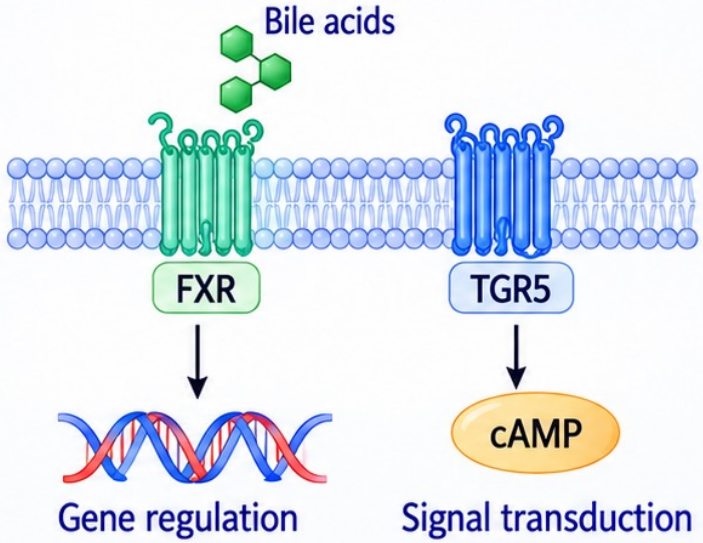
Pathway databases describe relationships between genes, proteins and molecules.

KEGG (pathway maps)

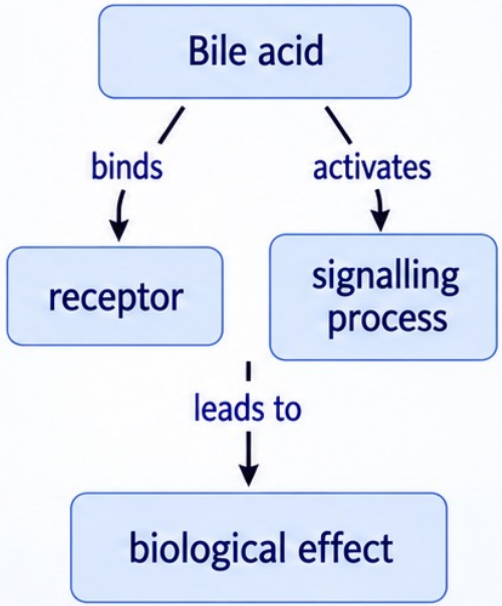


Reactome (detailed pathways)

Bile acid signalling pathway (example)



GO-CAM (causal model view)

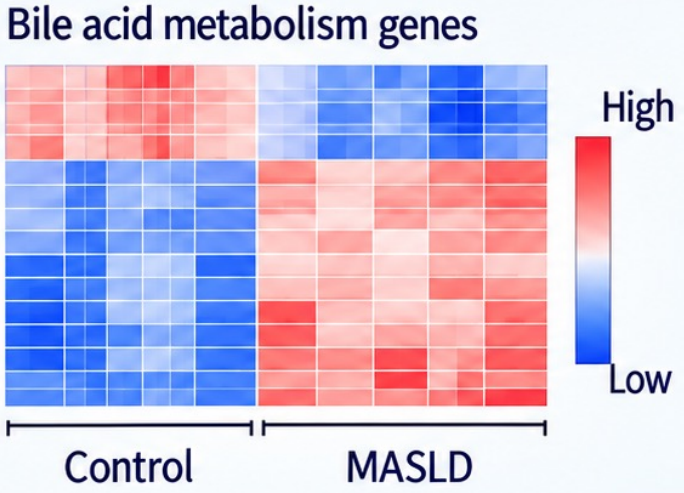


Take-home message: Pathway databases describe relationships, not measured pathway activity.

Enrichment Is Not Pathway Activation

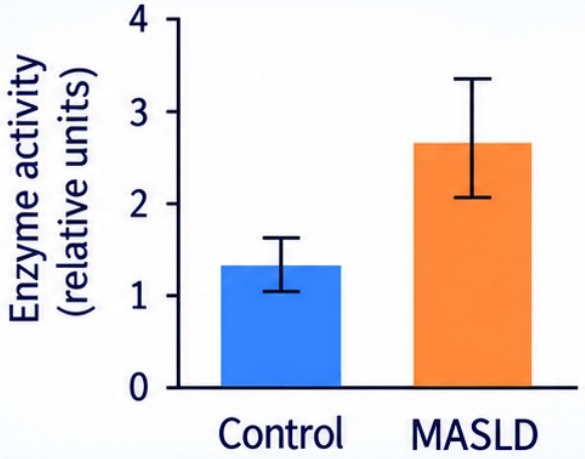
Expression enrichment does not necessarily indicate biochemical activity.

Gene expression enrichment



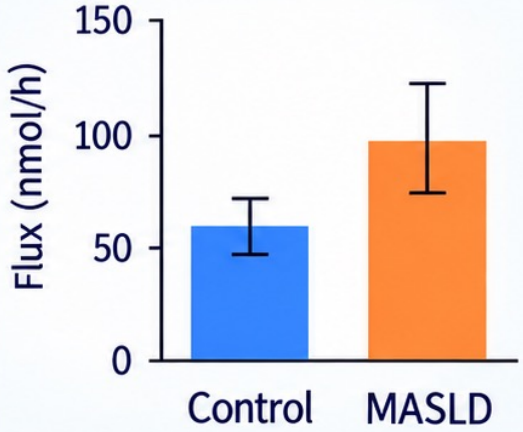
Enzyme activity

(measured protein function)



Metabolic flux

(biochemical outcome)



Take-home message: Expression enrichment alone does not establish pathway activation.

ORA: Are Selected Genes Overrepresented?

ORA tests whether a list of significant genes is enriched in a pathway.

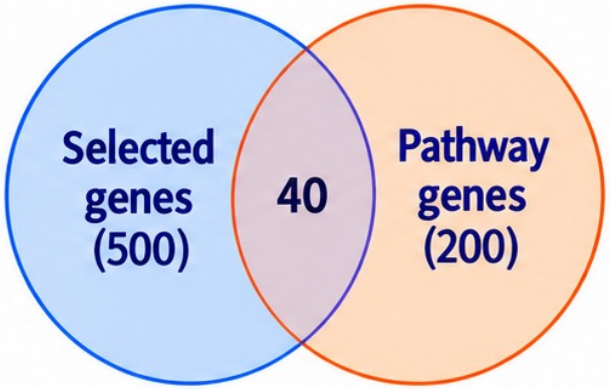
Input: selected gene list
(e.g. DE genes)

Gene1 Gene2 Gene3 Gene4

Gene5 Gene6 Gene7 Gene8

Gene9 Gene10 ...

Overlap with pathway genes



2 × 2 contingency table

	In pathway	Not in pathway	Total
In gene list	40	460	500
Not in list	160	9340	9500
Total	200	9800	10000

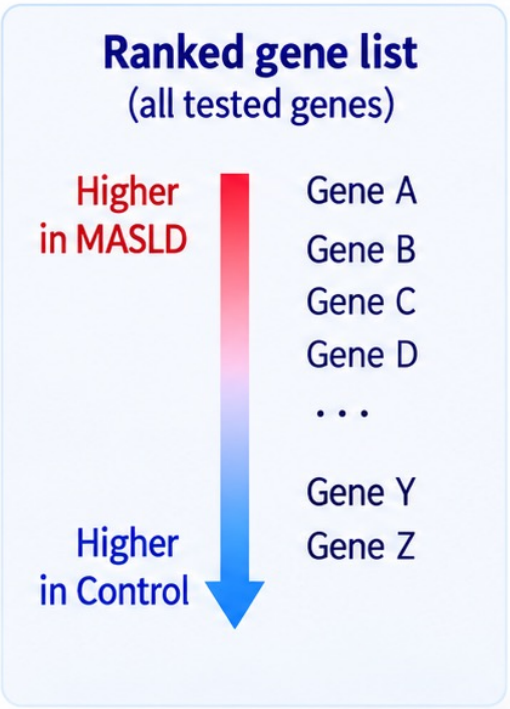
Statistical test

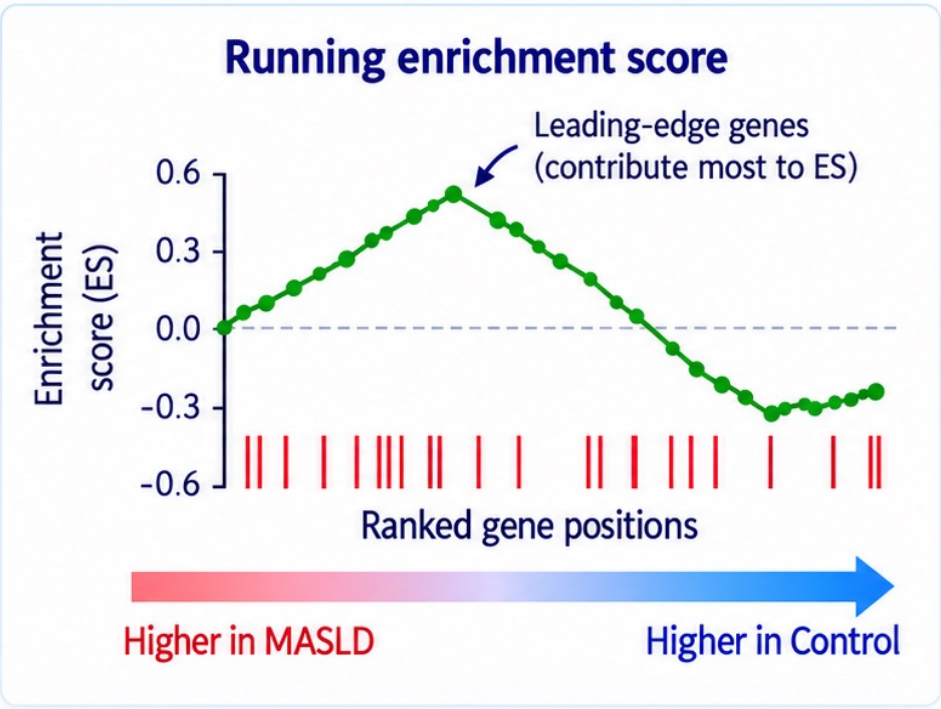
- Fisher's exact test / hypergeometric test
- Multiple testing correction (FDR)

Take-home message: ORA tests enrichment among a threshold-selected gene set.

GSEA: Are Pathway Genes Coordinately Ranked?

GSEA detects whether pathway genes show a coordinated ranking pattern.





Key outputs

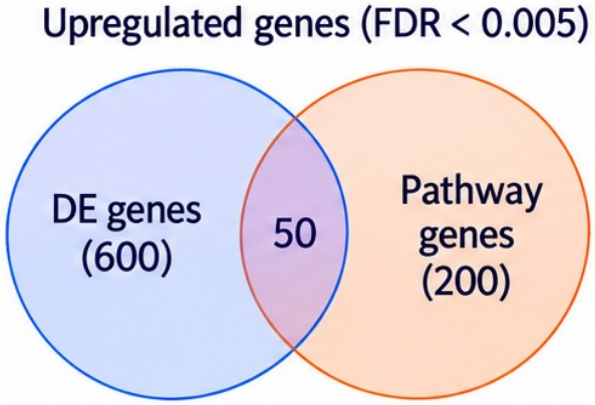
- Enrichment score (ES)
- Normalized ES (NES)
- P value and FDR
- Leading-edge genes

Take-home message: GSEA detects coordinated ranking patterns across all tested genes.

ORA Versus GSEA: Different Questions

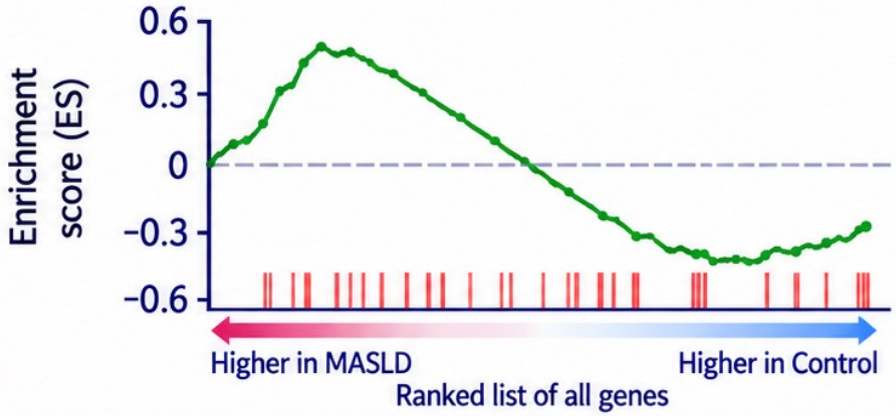
ORA and GSEA address different scientific questions using the same dataset.

ORA: enrichment among selected genes



Question:
Is the pathway overrepresented in the selected gene list?

GSEA: coordinated ranking of all genes



Question:
Do pathway genes tend to occur at the top or bottom of the ranked list?

Take-home message: The choice depends on the scientific question and input structure.

Gene IDs and Backgrounds Change Conclusions

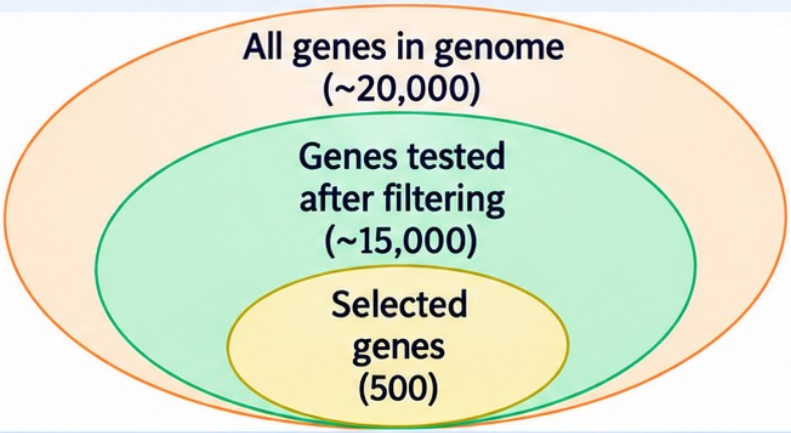
Correct ID mapping, gene universe and FDR control are essential.

ID mapping

Ensembl ID ENSG000001 → Mapping → Entrez ID 12345

- Use consistent ID type
- Check one-to-one mapping

Tested gene universe



Pathway enrichment workflow

Gene list (with IDs) → Select gene universe (tested genes) → Run ORA / GSEA (GO / KEGG / Reactome) → FDR control ($q < 0.05$) → Interpret results

Take-home message: ID mapping and background selection directly affect pathway results.

Connect Genomics, RNA and Proteins

From genetic variation to molecular function, with evidence at each step.

Genetic variant



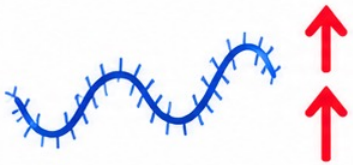
Risk variant
(eQTL)

Chromatin regulation



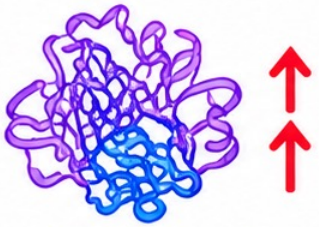
Enhancer /
promoter activity
(ATAC-seq / ChIP-seq)

Gene expression



mRNA level
(RNA-seq)

Protein abundance / function



Protein level
(proteomics)

↓

Protein function
(e.g. enzyme activity)

Potential regulation
(association)

Measured activity

Measured abundance

Measured function
(when available)

Take-home message: Regulatory-to-functional links require evidence at each step.

From Molecular Data to Biological Functions

Integrating multi-omics and multi-scale data to reveal biology.

Genomics
DNA variants,
regulatory elements

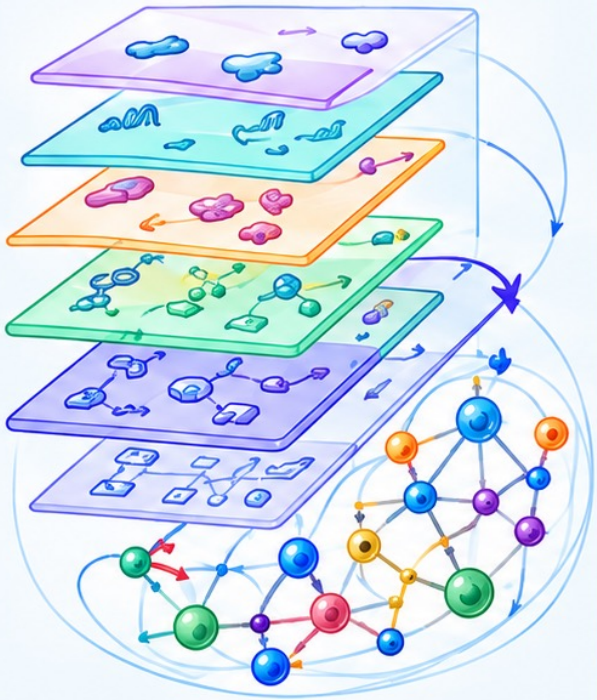
Transcriptomics
Gene expression,
splicing, non-coding RNA

Proteomics
Protein abundance,
PTMs, interactions

Metabolomics
Small molecules,
metabolic pathways

Microbiome
Bacteria, fungi, viruses,
metabolites

Multi-omics integration



Biological functions

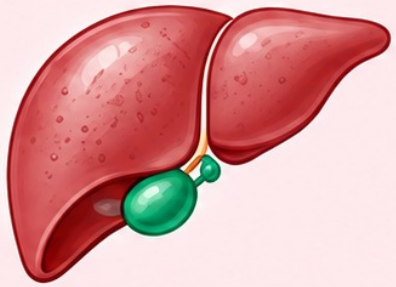
- Metabolism
- Immunity
- Signaling
- Tissue remodeling

Take-home message: Multi-omics integration bridges molecules to functions, revealing systems biology.

Four Host Functions Connect Multi-Omics

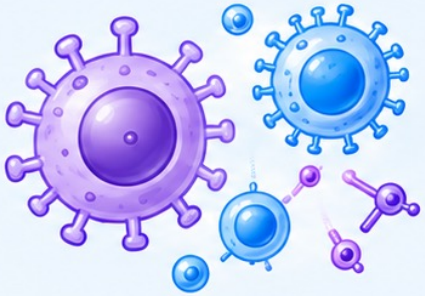
Different omics layers inform four key host functions in a coordinated manner.

Metabolism



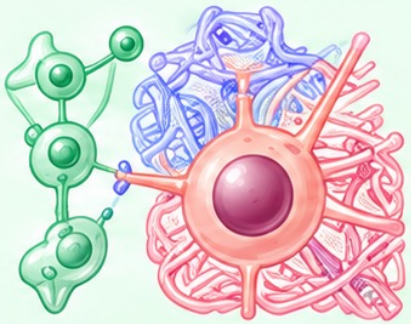
- Energy metabolism
- Lipid metabolism
- Glucose homeostasis
- Nutrient sensing

Immune response



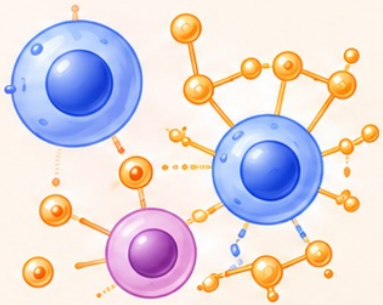
- Innate immunity
- Adaptive immunity
- Cytokine signaling
- Inflammation resolution

Tissue remodeling



- ECM remodeling
- Fibrosis
- Angiogenesis
- Wound healing

Cellular communication



- Cell-cell interactions
- Paracrine signaling
- Hormonal regulation
- Neural signaling

Take-home message: Multi-omics reveals how different host functions are coordinated and interlinked.

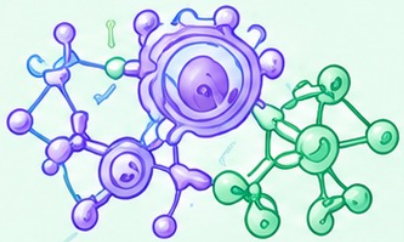
Gene Ontology: What Do Genes Do?

GO classifies gene functions into biological process, molecular function and cellular component.

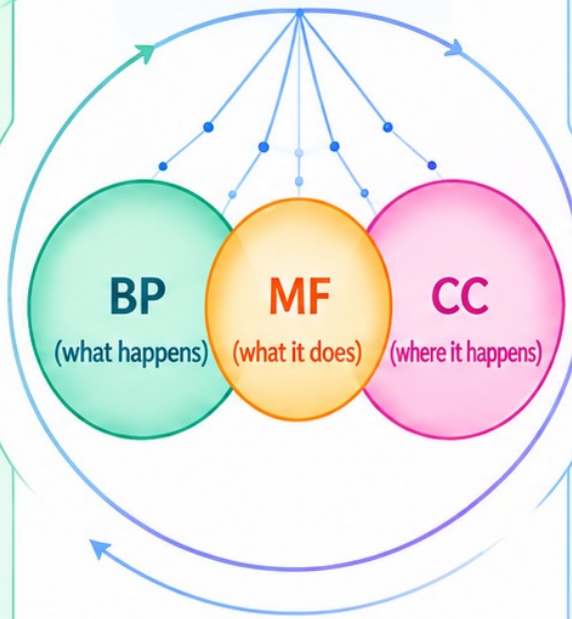
Gene Ontology (GO)

Biological Process (BP)

Biological objectives and pathways



- Immune response
- Lipid metabolism
- Inflammatory signaling
- Cell proliferation



Molecular Function (MF)

Molecular activities at the biochemical level



- Enzyme activity
- Receptor binding
- Transporter activity
- DNA / RNA binding

Cellular Component (CC)

Where gene products act in the cell



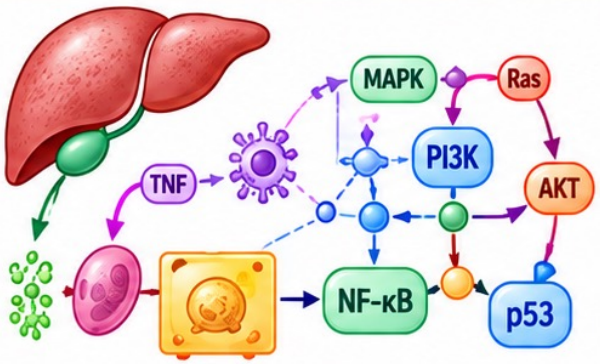
- Plasma membrane
- Mitochondrion
- Nucleus
- Extracellular region

Take-home message: GO provides a structured vocabulary to interpret gene functions.

KEGG and Reactome: Mapping Biological Systems

Pathway databases describe relationships between genes, proteins and molecules.

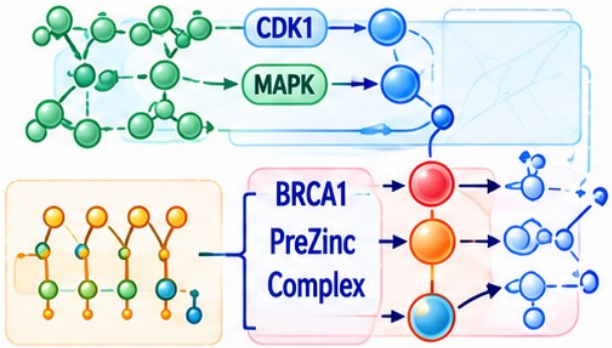
KEGG (pathway maps)



- Metabolic pathways
- Signaling pathways
- Disease pathways
- Drug targets

KEGG

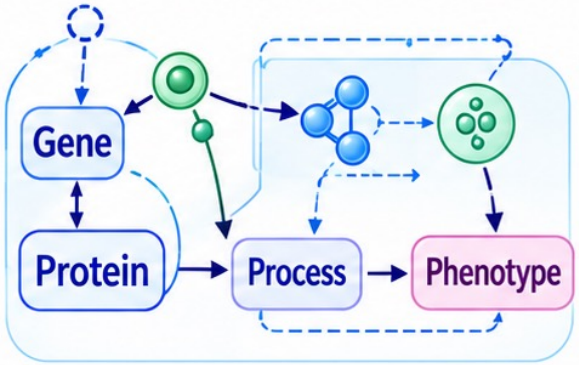
Reactome (detailed pathways)



- Molecular reactions
- Pathway hierarchy
- Disease pathways
- Cross-species

reactome

GO-CAM (causal model view)



- Causal relationships
- Evidence-based
- Biological context

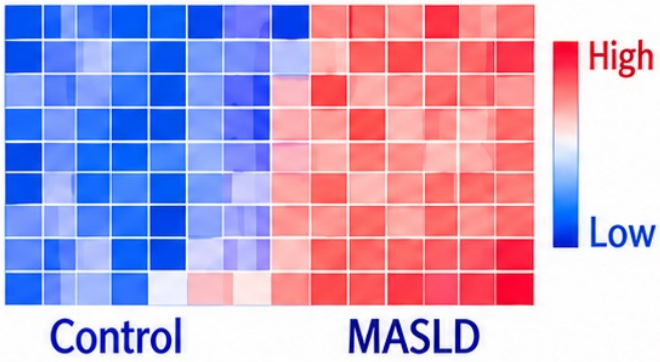
GO-CAM

Take-home message: Pathway databases turn gene lists into biological context and systems.

Enrichment Is Not Pathway Activation

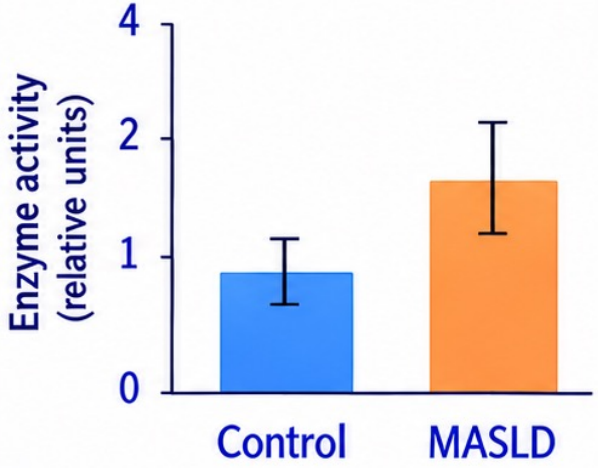
Expression enrichment does not necessarily indicate biochemical activity.

Gene expression enrichment
(e.g. MASLD vs Control)

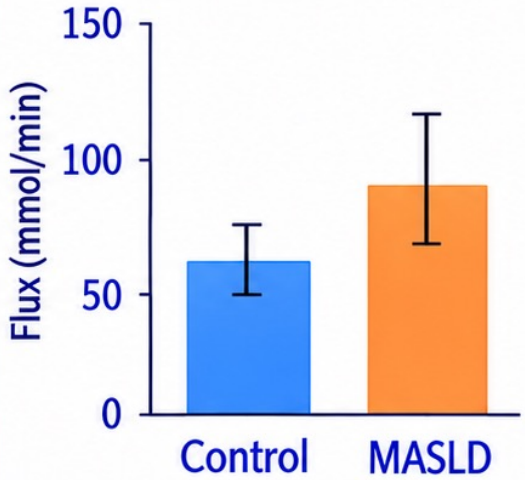


- Genes are over-represented in a pathway
- Does not mean the pathway is activated

Enzyme activity
(measured protein function)



Metabolic flux
(biochemical outcome)



Take-home message: Enrichment suggests association, not direct functional activation.

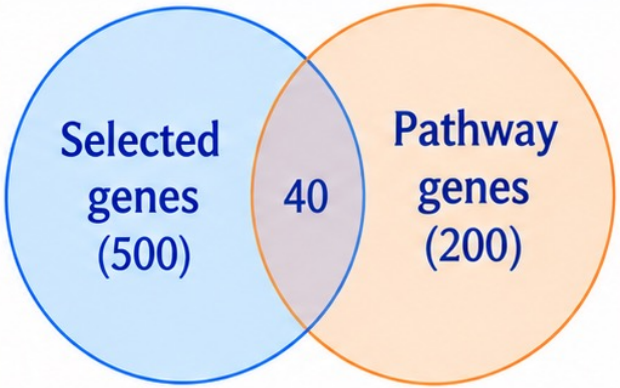
ORA: Are Selected Genes Overrepresented?

ORA tests whether a list of significant genes is enriched in a pathway.

Input: selected gene list
(e.g. DE genes)

Gene1 Gene2 Gene3 Gene4
Gene5 Gene6 Gene7 Gene8
Gene9 Gene10 ...

Overlap with pathway genes



2 × 2 contingency table

	In pathway	Not in pathway	Total
In gene list	40	460	500
Not in list	160	9340	9500
Total	200	9800	10000

Statistical test

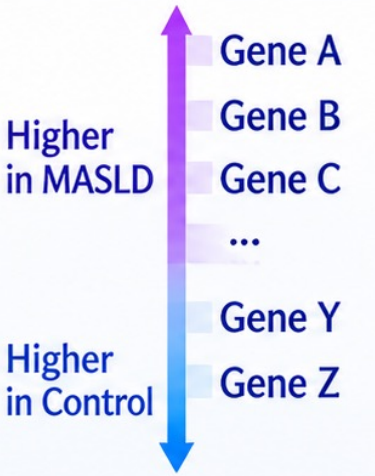
- Fisher's exact test / hypergeometric test
- Multiple testing correction (FDR)

Take-home message: ORA tests enrichment among a threshold-selected gene set.

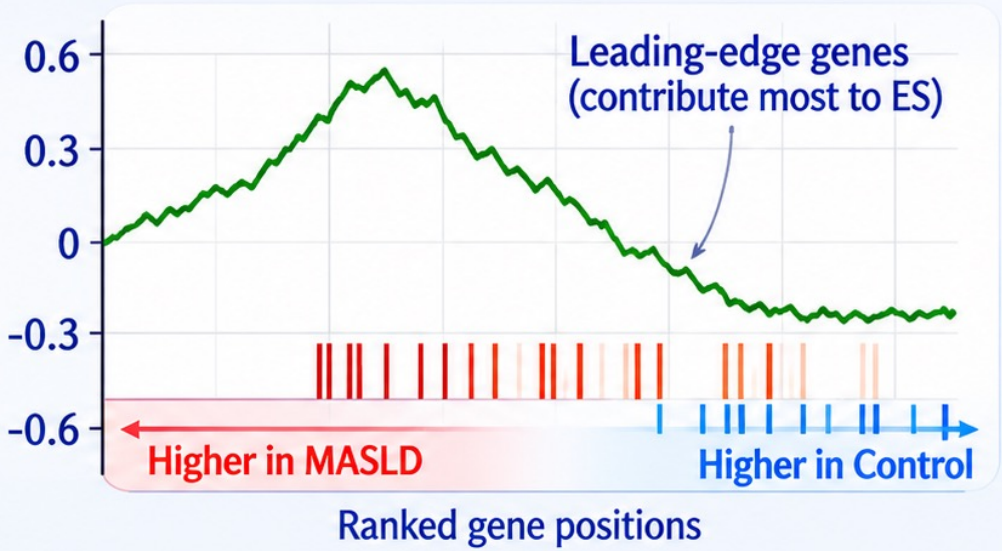
GSEA: Are Pathway Genes Coordinated Ranked?

GSEA detects whether pathway genes show a coordinated **ranking pattern.**

Ranked gene list
(all tested genes)



Running enrichment score



Key outputs

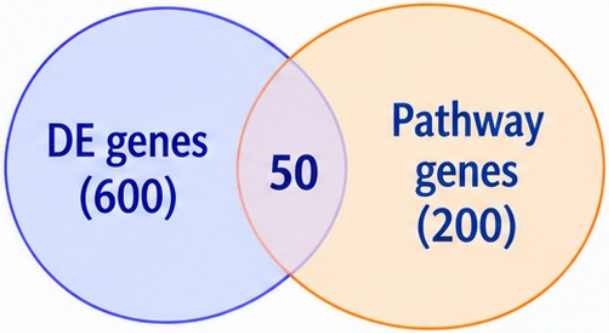
- Enrichment score (ES)
- Normalized ES (NES)
- P-value and FDR
- Leading-edge genes

Take-home message: GSEA detects coordinated ranking patterns across all tested genes.

ORA Versus GSEA: Different Questions

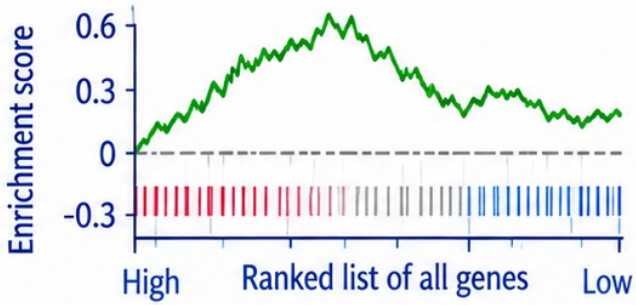
ORA and GSEA address different scientific questions using the same datasets.

ORA: enrichment among selected genes



Question:
Is the pathway overrepresented in the selected gene list?

GSEA: coordinated ranking of all genes



Question:
Do pathway genes tend to occur at the top or bottom of the ranked list?

ID map/id bal. | Run ORA / GSEA

- Ensembl ID/Coord. Loci → Run ORA / GSEA
- One-to-one mapping → GO / KEGG / Reactome
- Reactome → Reactome
- 115 seq → MSigDB (C2/C5)

Pathway enrich workflow

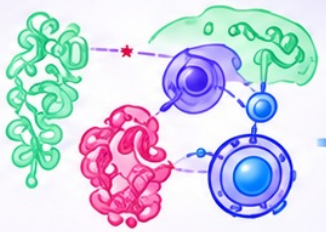
Gene list (with IDs) → GO/Reactome (optional) → Run ORA / GSEA (q < 0.05)

Take-home message: The choice depends on the scientific question and input gene structure.

Gene IDs and Backgrounds Change Conclusions

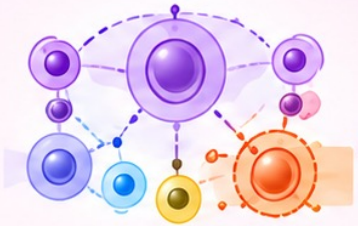
From genetic mapping, gene universe function, with evidence evidence at each step.

Genetic variant



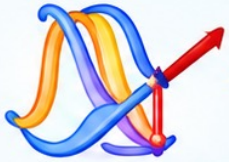
- Risk variant (eQTL)
- Enhancer / promoter
- (q < 0.05)

Tested gene universe



- Proper, broad gene universe
- Proper gene testing
- Focal/selected genes

ID mapping / Inference functions



Risk variant (eQTL)

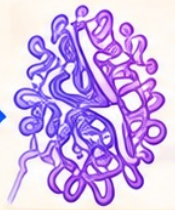
Chromatin regulating



- Tss/pos
- mRNA level (RNA-seq)

- Enhancer / promoter activity (ATAC-seq / ChIP-seq)

Protein abundance / function



- Function
- abundance / function

- Protein level (e.g. enzyme activity)

Potential regulation **(association)** → Measured activity → Measured abundance → Measured function **(often available)**

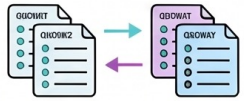
Take-home message: Regulatory-to-functional links require evidence at each step.

Multi-Omics Data Integration: From Audit to Biological Evidence



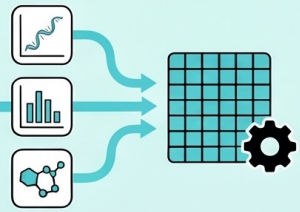
1. Sample Auditing & Strategy Selection

Audit biological units and match donor keys across layers



Ensure unique donor mapping and control for batch covettules before running cross-omics models

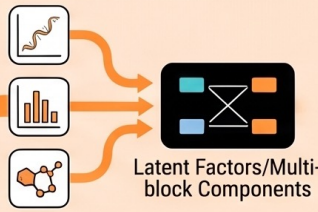
EARLY INTEGRATION



What is Joined: Scaled features before model fitting

Primary Failure Mode: Feature block dominance and severe overtitting

INTERMEDIATE INTEGRATION



What is Joined: Latent factors or multi-block components

Primary Failure Mode: Inferred factors tracking technical batch effects

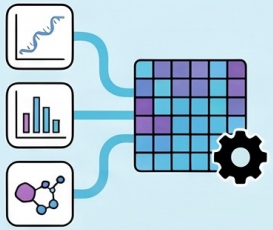
LATE INTEGRATION



What is Joined: Evidence derived post layer-specific analysis

Primary Failure Mode: Fails to model direct cross-assay feature covariance

EARLY INTEGRATION



Separate unsupervised factor discovery from supervised signatures

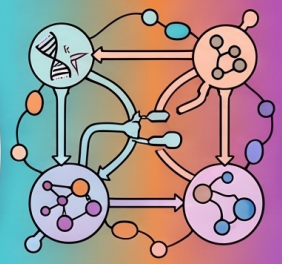
(Apply MOFA2 for unsupervised exploratory axes and nested-validated DIABLO for supervised signatures)

Match enrichment design to specific statistical questions



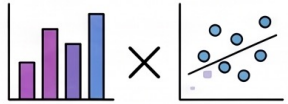
ORA
tests thresholded genes against a background universe

GSEA
requires signed gene rankings



2. Pathway Analysis & Cross-Layer Evidence Synthesis

Keep non-commensurate quantitative scales strictly separate



RNA enrichment scores

Metabolite impact values

Construct qualitative evidence matrices to show convergence

	Pathway A	Pathway B	Pathway C
Layer 1	✓	✓	✓
Layer 2	✓	○	○
Layer 3	✓	○	✓

Summarize cross-layer concordance as observational support without overclaiming mechanistic causality.